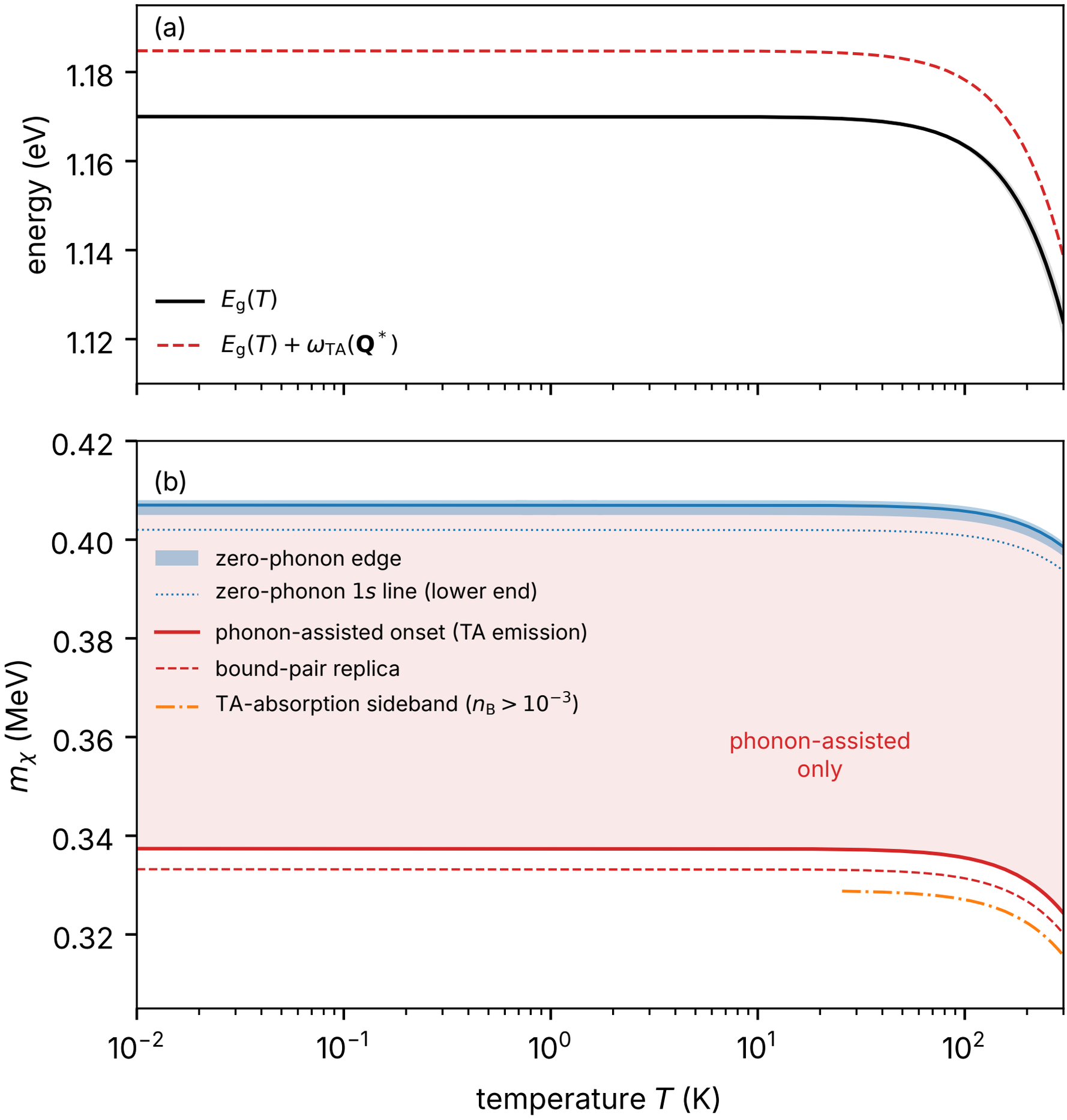
(a)
energy (eV)
1.18
1.16
1.14
1.12
$E_g(T)$
$E_g(T) + \omega_{TA}(\mathbf{Q}^*)$
(b)
m_χ (MeV)
0.42
0.40
0.38
0.36
0.34
0.32
zero-phonon edge
zero-phonon 1s line (lower end)
phonon-assisted onset (TA emission)
bound-pair replica
TA-absorption sideband ($n_B > 10^{-3}$)
phonon-assisted only
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
temperature T (K)